

Tasks 1A–4 and Extensions: Exploratory Data Analysis and Modelling of Wine Quality

Sofia French

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Overview

This document presents the full exploratory data analysis (EDA) and modelling tasks conducted as part of the wine quality prediction project. Both red and white wine datasets are analysed. Relationships between alcohol, residual sugar, and quality are explored, followed by feature engineering and classification/regression modelling. Extensions investigate interactions between alcohol and sweetness.

Task 1A: Initial Data Analysis

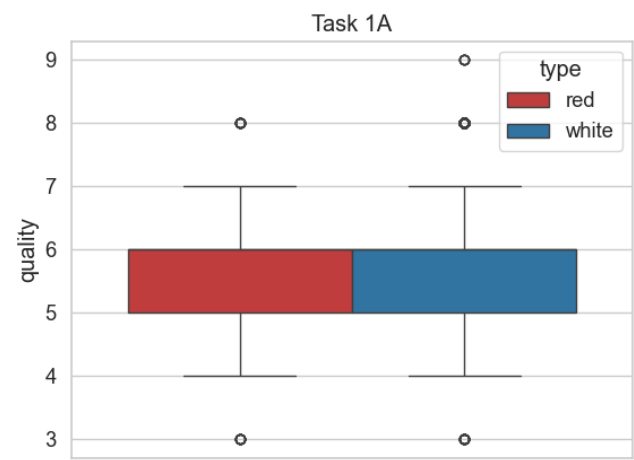


Figure 1: Boxplot of Quality Scores in Red and White Wine Datasets

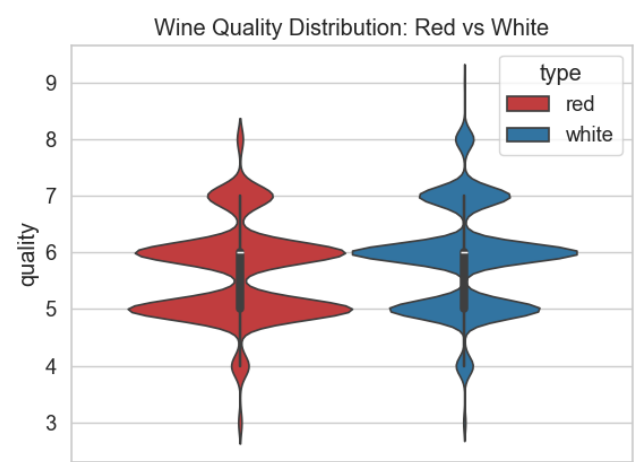


Figure 2: Violin Plot: Red vs White Wine Quality Distribution

Interpretation: Most red and white wines are rated between 5 and 6, but white wines have a wider spread. This may reflect differences in production or grape characteristics.

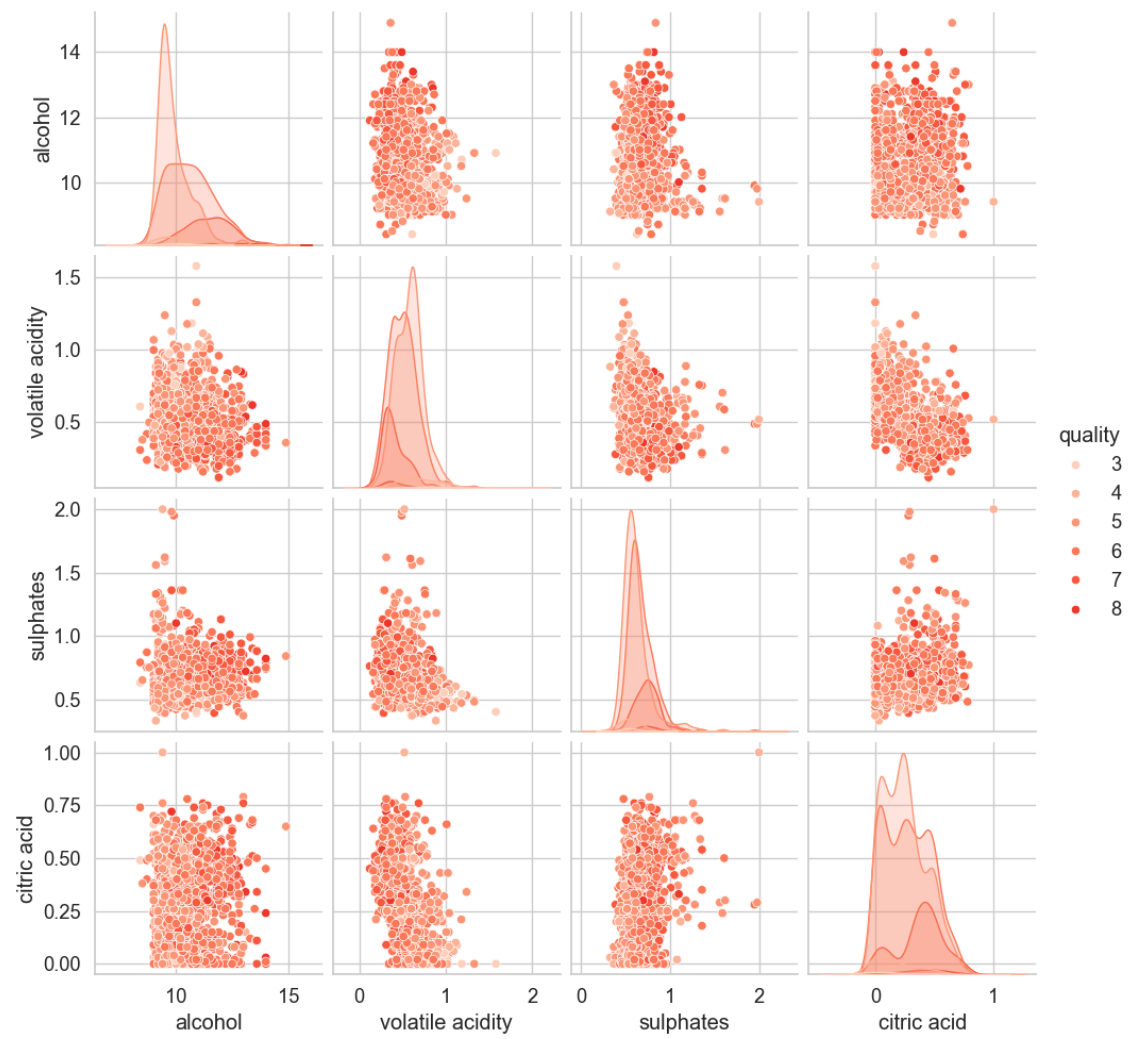


Figure 3: Red Wine: Scatterplot Matrix of Selected Features by Quality

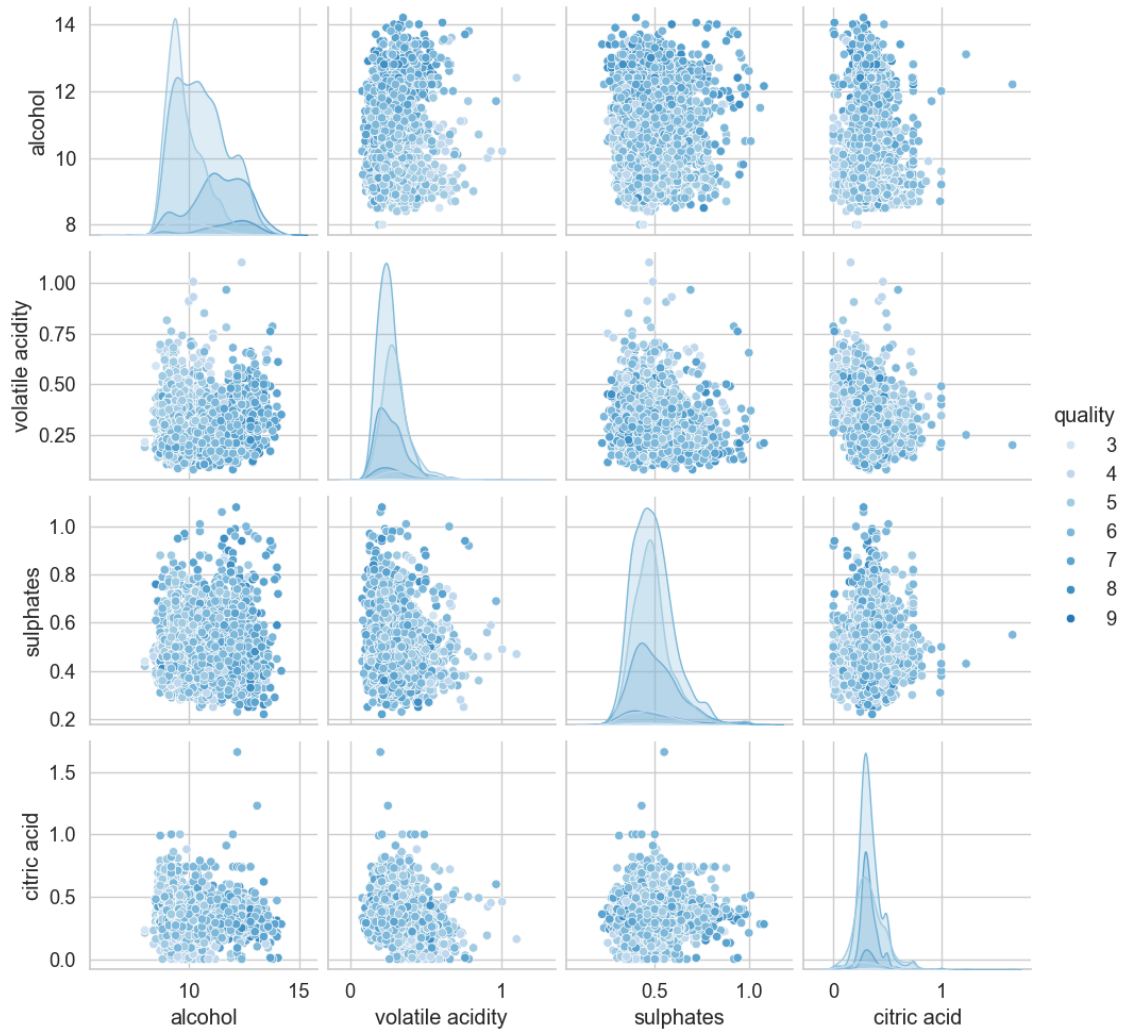


Figure 4: White Wine: Scatterplot Matrix of Selected Features by Quality

Task 1B: Alcohol Content and Quality

Alcohol content was categorised as low, mid, or high based on the mean and standard deviation.

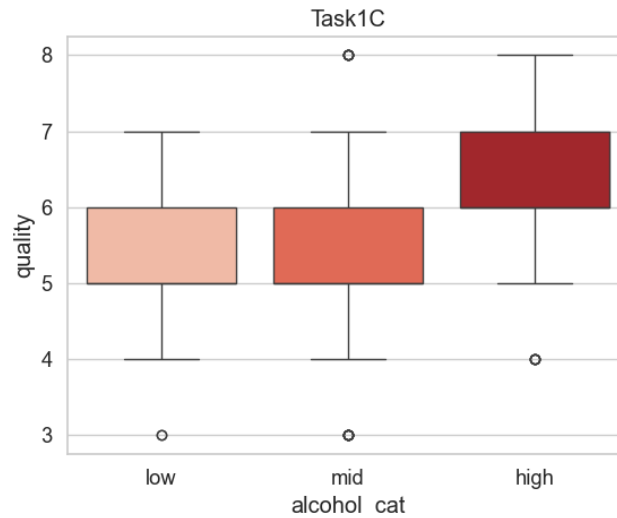


Figure 5: Red Wine: Quality by Alcohol Category

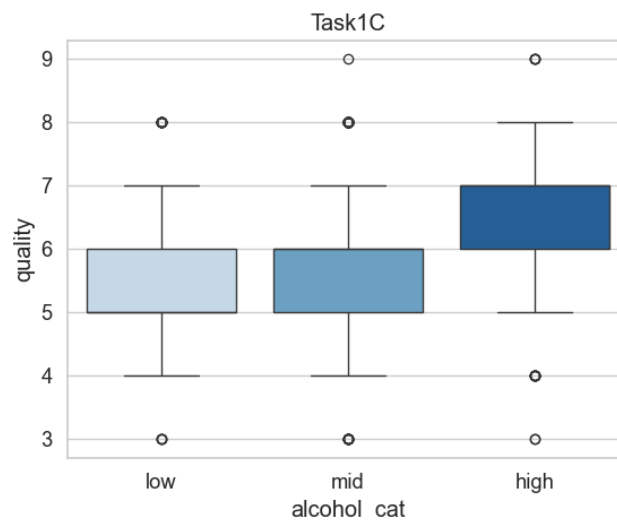


Figure 6: White Wine: Quality by Alcohol Category

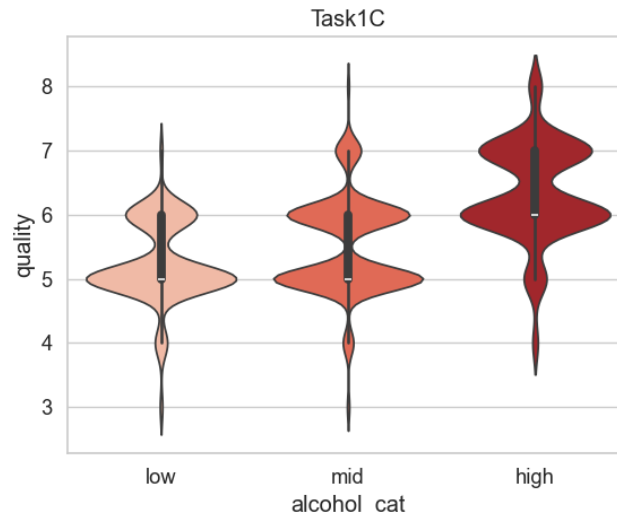


Figure 7: Red Wine: Violin Plot of Quality by Alcohol Category

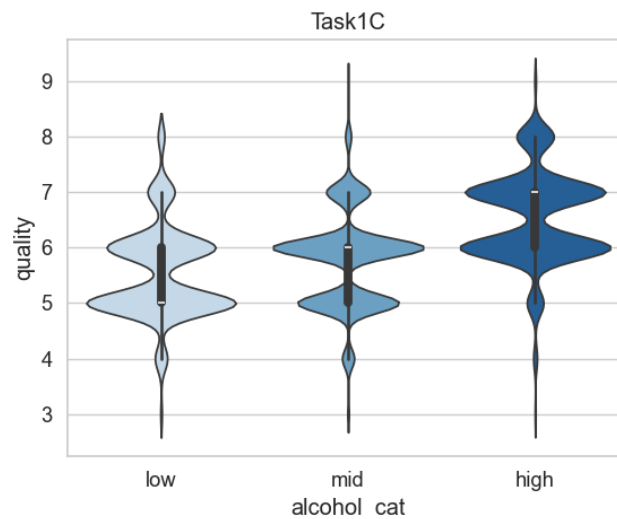


Figure 8: White Wine: Violin Plot of Quality by Alcohol Category

Conclusion: Alcohol is positively associated with quality in both datasets, with red wine showing a clearer pattern.

Task 1C: Residual Sugar and Sweetness

The feature `isSweet` was created by splitting at the median of residual sugar.

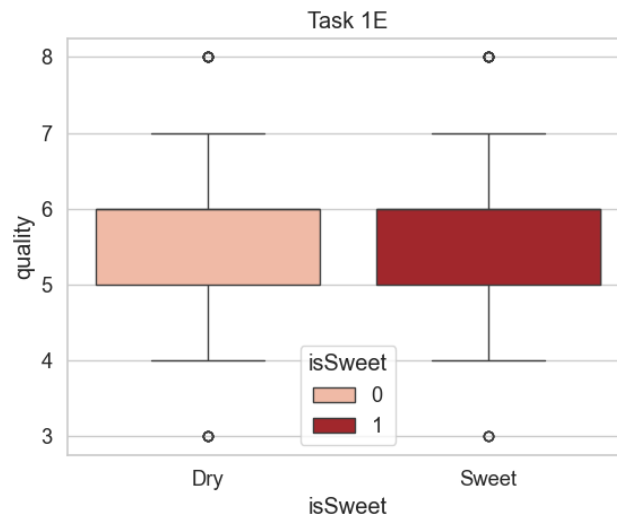


Figure 9: Red Wine: Quality by Sweetness (`isSweet`)

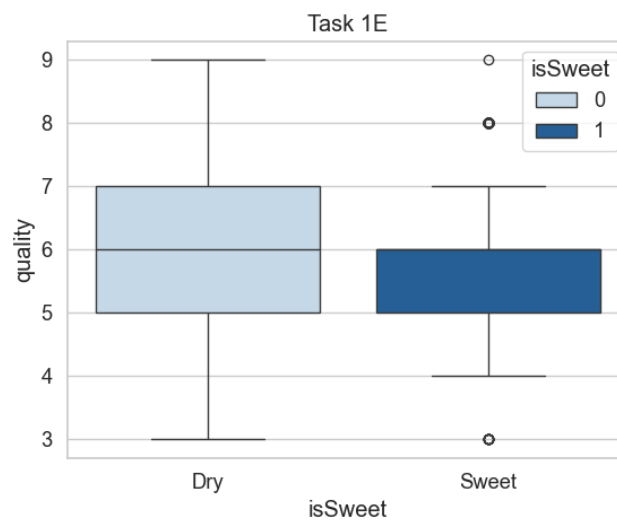


Figure 10: White Wine: Quality by Sweetness (`isSweet`)

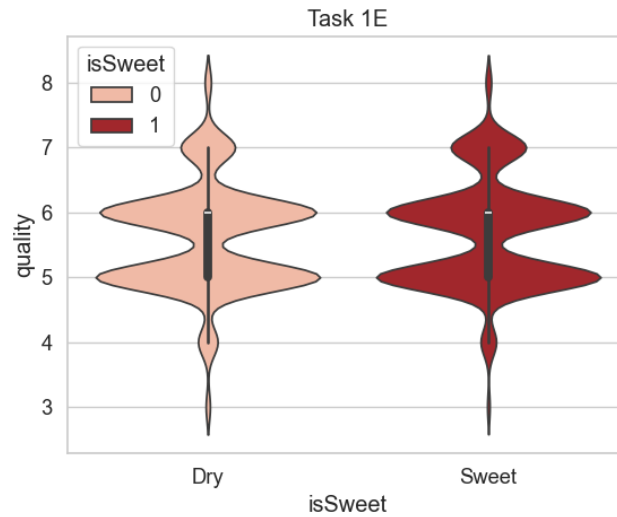


Figure 11: Red Wine: Violin Plot of Quality by Sweetness

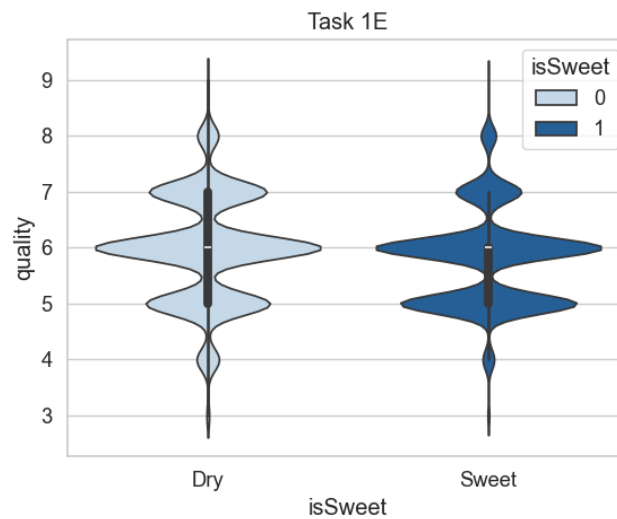


Figure 12: White Wine: Violin Plot of Quality by Sweetness

Conclusion: Sweetness has a weak negative effect on quality, especially in white wine.

Extension 1: Alcohol & Sweetness Interaction

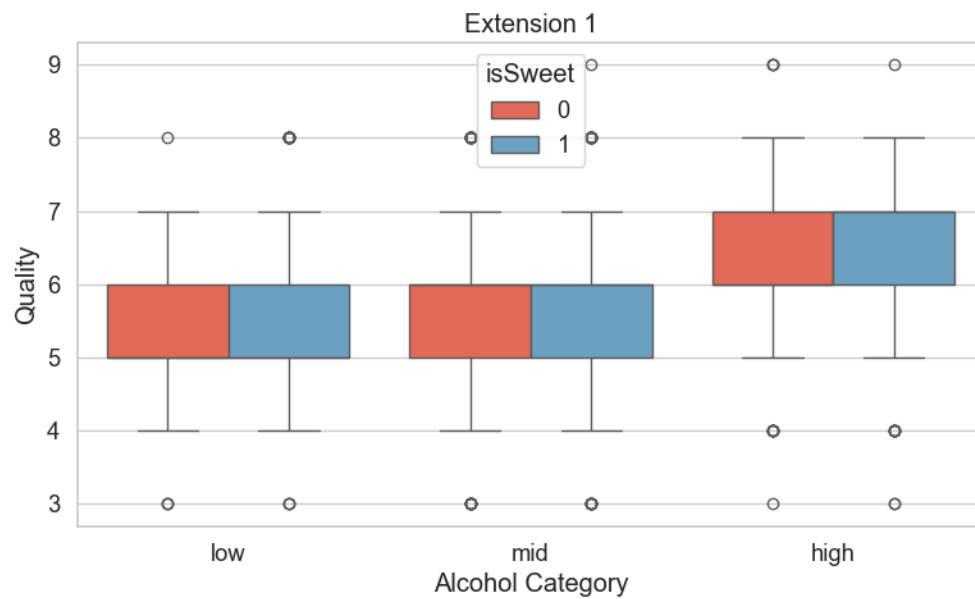


Figure 13: All Wines: Quality by Alcohol and Sweetness

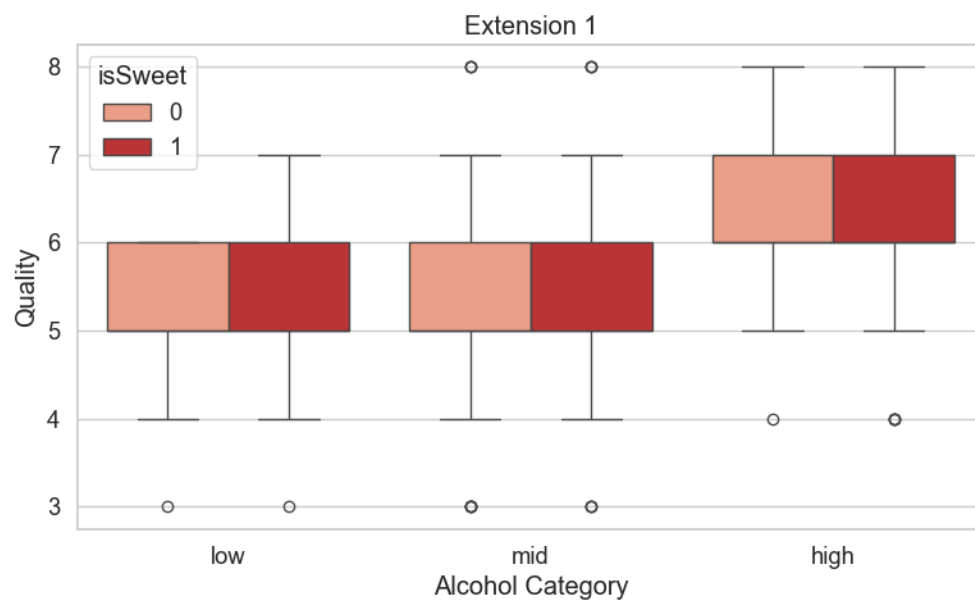


Figure 14: Red Wine: Quality by Alcohol and Sweetness

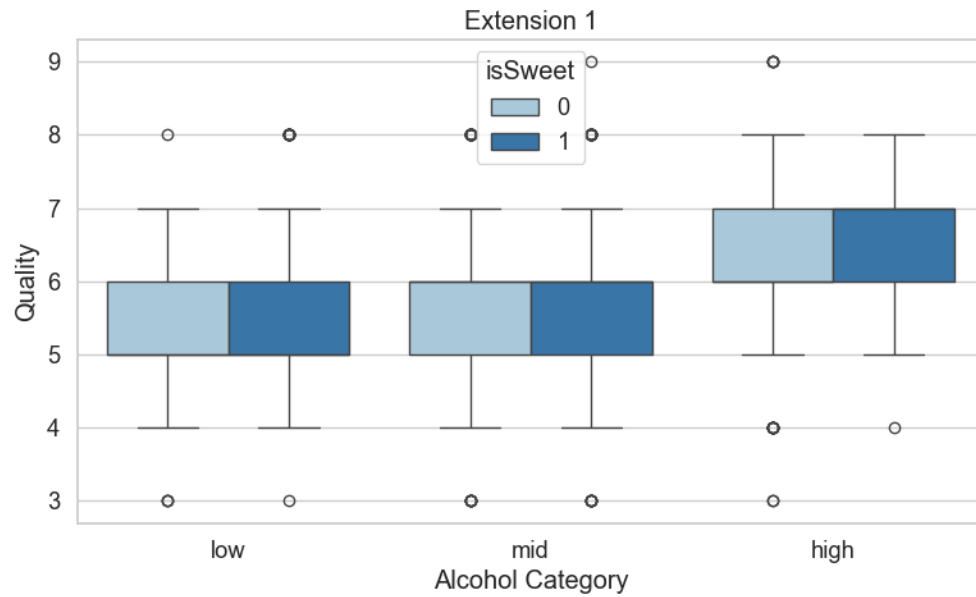


Figure 15: White Wine: Quality by Alcohol and Sweetness

Interpretation: Quality increases with alcohol regardless of sweetness, but dry wines tend to benefit more. Sweetness reduces average quality in high-alcohol white wines.

Task 2: Correlation Analysis

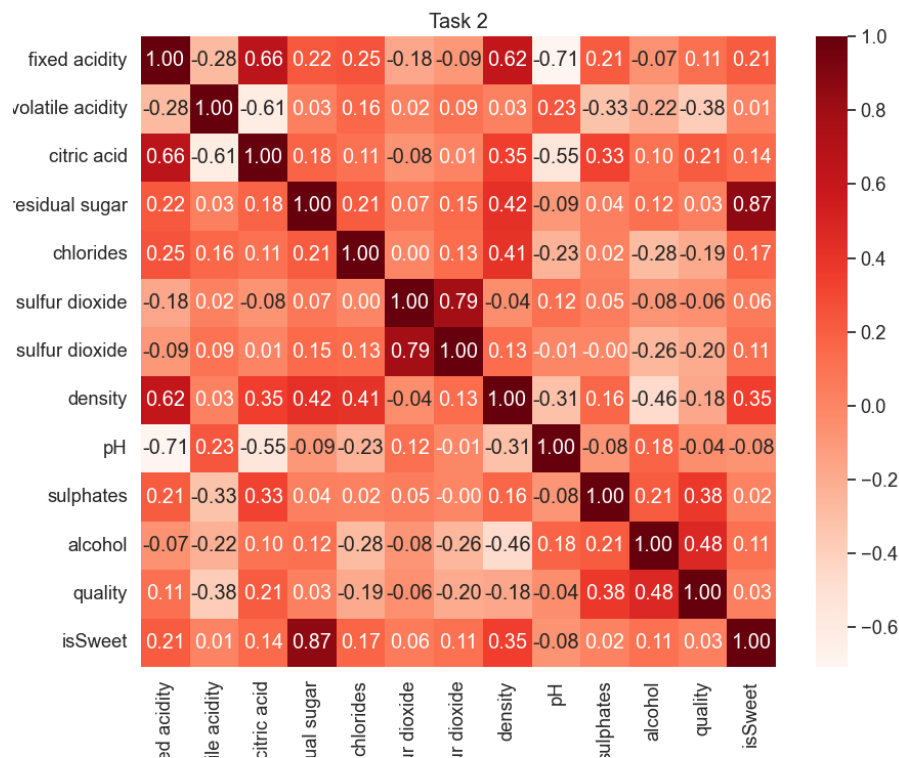


Figure 16: Red Wine: Pearson Correlation Matrix

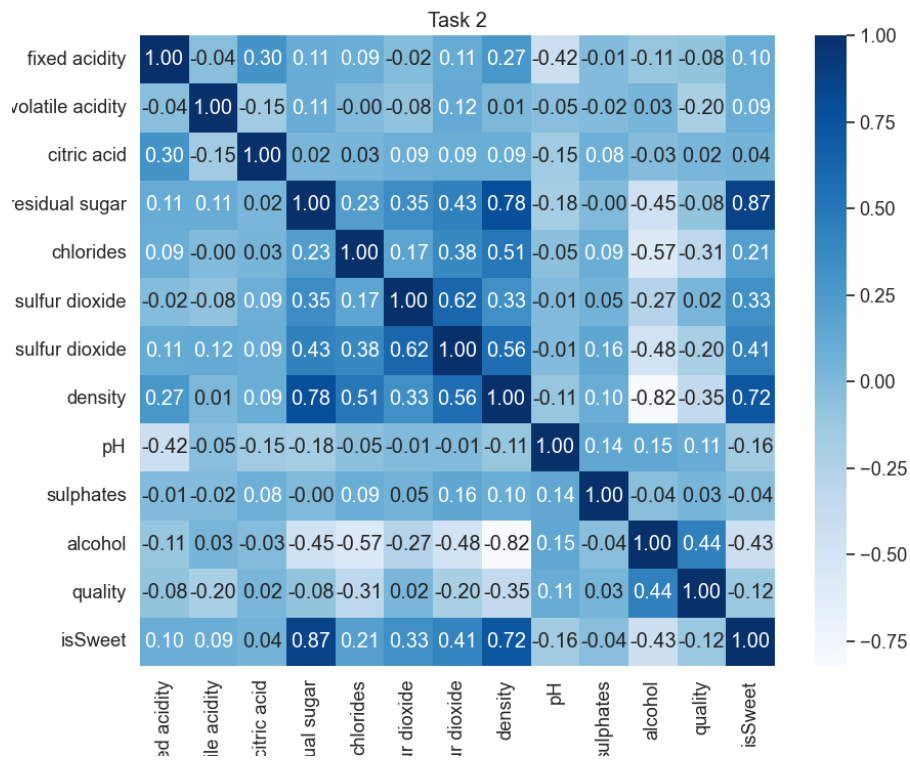


Figure 17: White Wine: Pearson Correlation Matrix

Task 3: Binary Classification

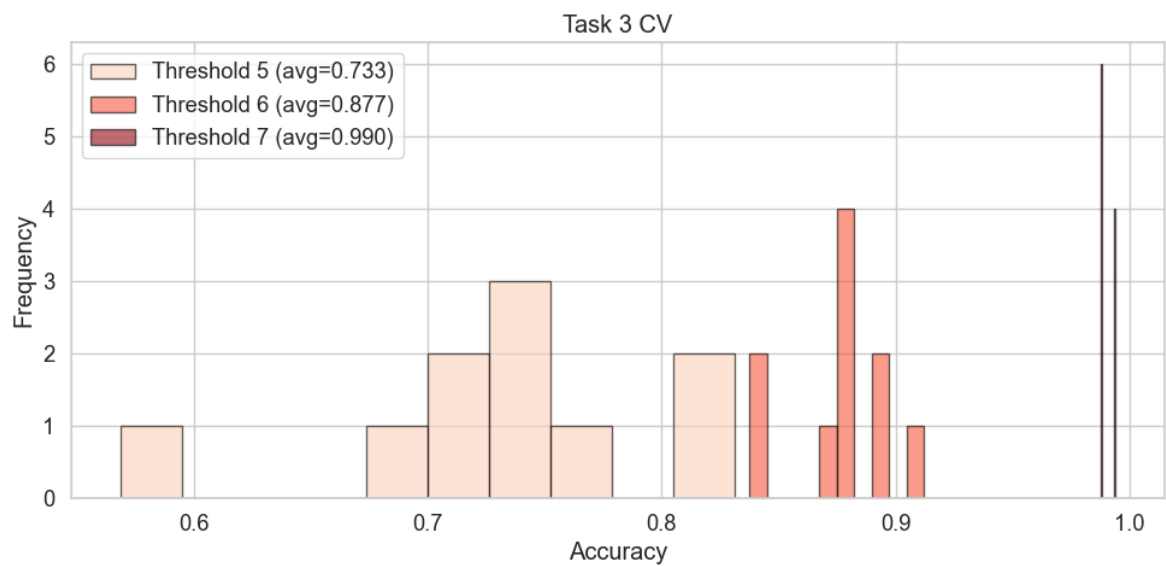


Figure 18: Red Wine: Cross-Validation Accuracy Across Thresholds

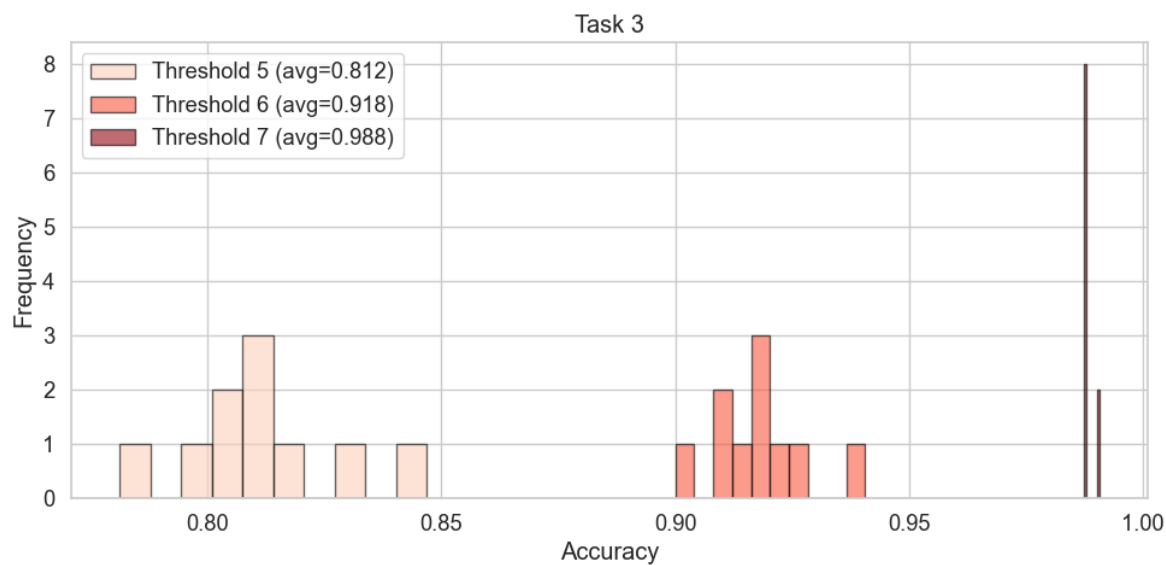


Figure 19: Red Wine: Test Set Accuracy Distribution

Conclusion: Threshold 6 yielded the best balance of performance and class distribution.

Task 4: Regression and ROC Evaluation

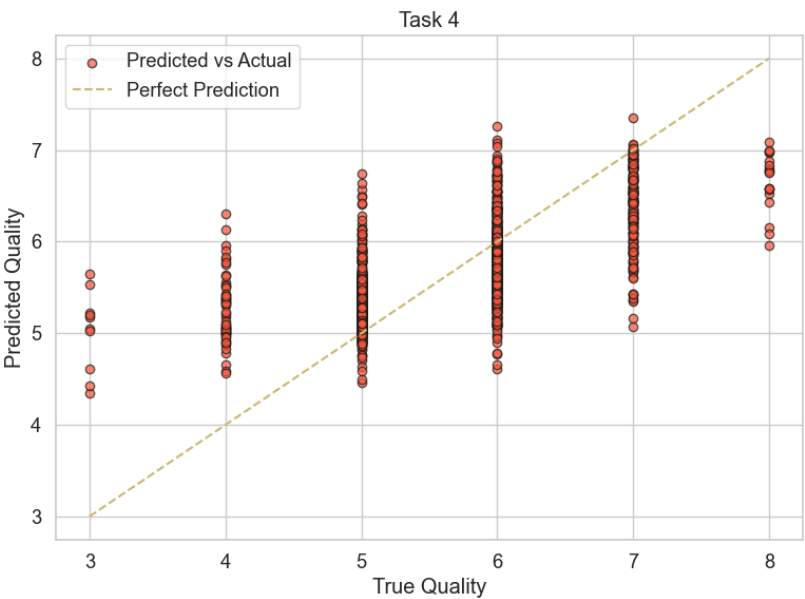


Figure 20: Red Wine: Predicted vs Actual Quality (Regression)

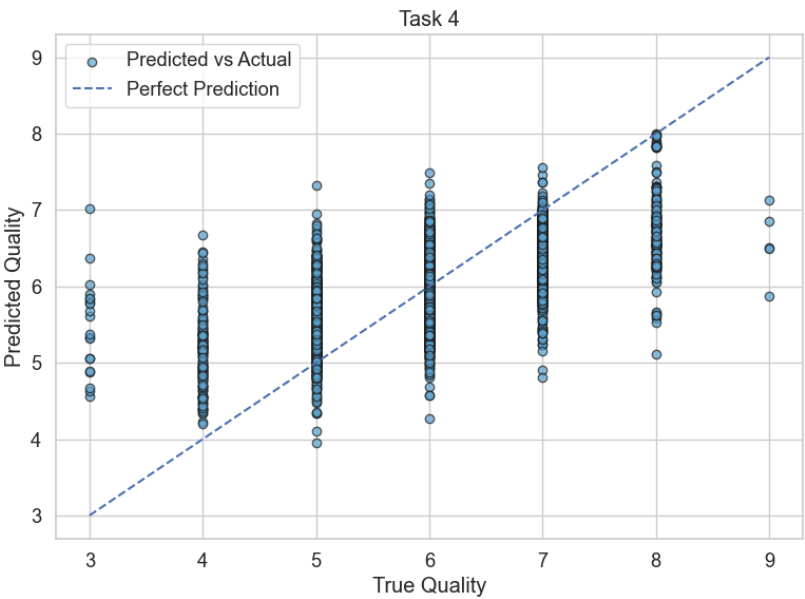


Figure 21: White Wine: Predicted vs Actual Quality (Regression)

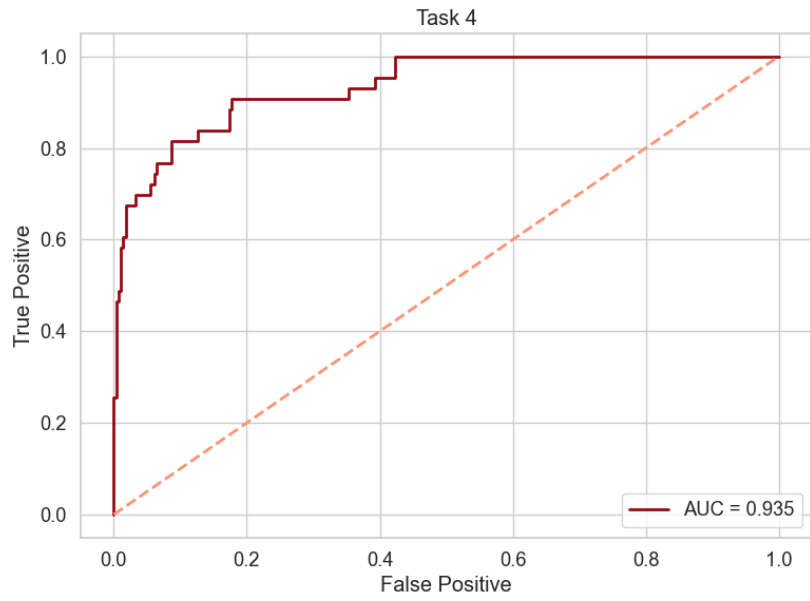


Figure 22: Red Wine: ROC Curve for Threshold 6

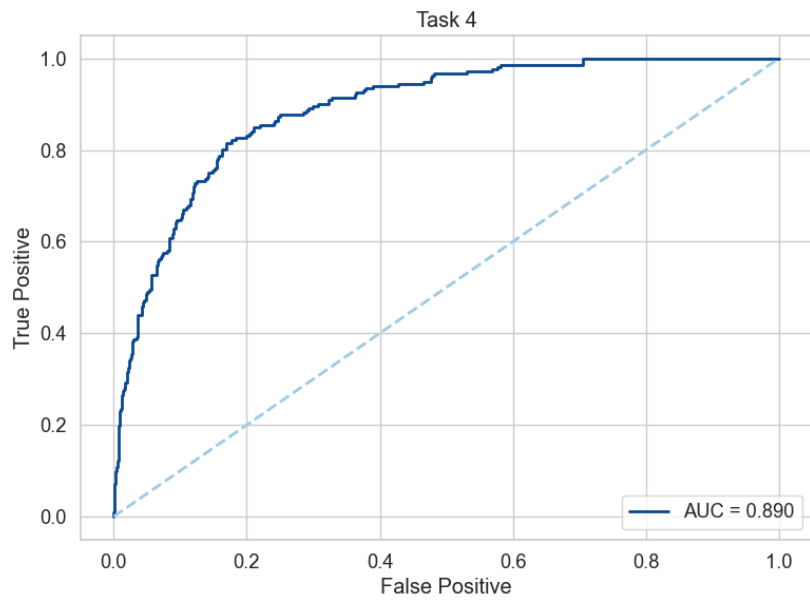


Figure 23: White Wine: ROC Curve for Threshold 6